Dual-Domain Aligned Temporal–Spatial–Spectral Fusion Networks for No-Paired Hyperspectral and Multispectral Images

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***Abstract*— The objective of temporal–spatial–spectral fusion for remote sensing images is to reconstruct imagery with enhanced temporal, spatial, and spectral resolution. This process aims to integrate the rich spectral information inherent in hyperspectral images (HSIs) with the temporal and spatial details provided by multispectral images (MSIs). However, existing methodologies predominantly depend on the availability of strictly co-registered HSI-MSI pairs, constraining their practical applicability due to the significant challenge of simultaneously acquiring such paired data over the same geographical region. Furthermore, these methods frequently encounter difficulties in effectively harmonizing the inherent radiometric and spatial discrepancies between multisource remote sensing images, often resulting in suboptimal fusion results. To address these limitations, we propose a novel dual-domain aligned temporal–spatial–spectral fusion network (DAFN) designed for unpaired hyperspectral image (HSI) and multispectral image (MSI) imagery. DAFN integrates a style transfer module to mitigate radiometric inconsistencies, a spatial alignment module to rectify geometric misregistrations, dedicated modules for change feature extraction and spatial information refinement, culminating in feature fusion to accurately reconstruct high temporal–spatial resolution hyperspectral imagery. Extensive experiments on both simulated and real-world datasets demonstrate that DAFN consistently outperforms state-of-the-art methods.**

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***Index Terms*— Domain alignment, hyperspectral, image fusion, multispectral.**

I. Introduction

HYPERSPECTRAL sensors typically capture a spectral range of 400–2500 nm, acquiring more than 100 spectral bands with fine spectral resolution [1]. Due to their exceptional target recognition capabilities, hyperspectral images (HSIs) have found widespread applications in fields such as environmental monitoring, resource exploration, and agricultural surveillance [2], [3]. However, because of the limited surface energy captured by the narrow spectral channels, HSIs generally exhibit low spatial resolution, with satellite revisit intervals often exceeding 30 days [4], [5]. Notable examples include the ZY-1 02D and GF-5B hyperspectral satellites from China, as well as the PRISMA hyperspectral satellite from Italy [6], [7]. In contrast, multispectral satellites offer superior spatial resolution, typically under 15 m, and higher temporal resolution, with revisit periods of less than ten days [8], [9]. Multispectral images (MSIs) provide essential complementary information that can enhance the resolution of HSIs [10]. Through image fusion, both the temporal and spatial resolution of HSIs can be effectively improved [11]. Current approaches for enhancing HSI resolution can generally be classified into two categories: image fusion methods and spectral super-resolution techniques, depending on whether auxiliary data are utilized [12], [13].

Image fusion methods primarily aim to enhance the resolution of HSI using MSI. Spectral super-resolution methods primarily aim to reconstruct HSI by establishing transformation mechanisms between HSI and MSI. Recent studies on HSI reconstruction have demonstrated the satisfactory performance of these methods [14], [15]. However, the following conditions hold.

1) Spectral super-resolution methods directly establish a transformation relationship between images. While these methods significantly improve spatial and temporal resolution, they still suffer from persistent spectral distortion. In contrast, image fusion methods, which utilize auxiliary information to enhance resolution, effectively reduce reconstruction errors.
2) Temporal–spatial–spectral fusion aims to simultaneously enhance the temporal and spatial resolution of HSI, with deep learning algorithms demonstrating considerable

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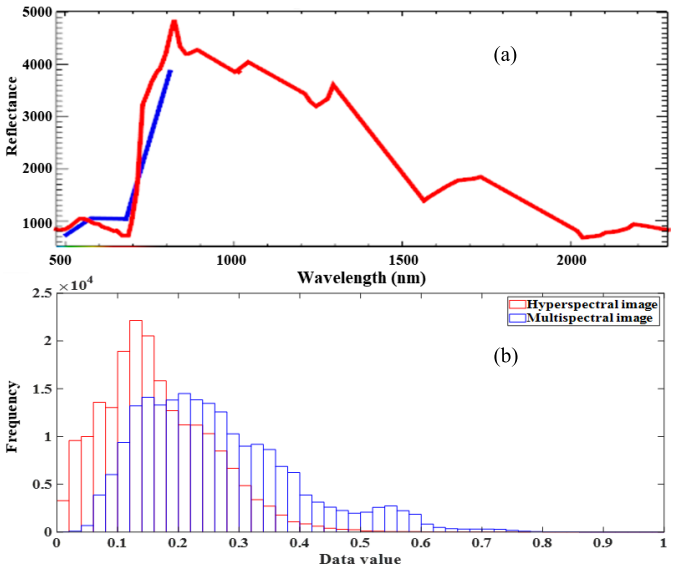


Fig. 1. Radiation differences between HSI and MSI. (a) Inconsistent spectral range. (b) Inconsistent distribution statistics.



Fig. 2. Spatial location differences between HSI and MSI (the arrow indicates the direction of offset, and the larger the size of the circle, the more severe the offset).

potential in this area. However, these methods typically require paired HSI and MSI from the same region at the same time.

3) There are discrepancies in spectral ranges and grayscale distributions between HSI and MSI (radiometric discrepancies) (see Fig. 1). Additionally, substantial spatial misalignments arise from differences in imaging platforms (see Fig. 2). Nevertheless, current temporal–spatial–spectral fusion methods have not sufficiently addressed these radiometric and spatial discrepancies, leading to suboptimal fusion results.

To address these challenges, we propose a dual-domain aligned temporal–spatial–spectral fusion network (DAFN) for unpaired HSI and MSI. The objective is to reconstruct high-spatial-resolution HSI at time (T) 2 using low-spatial-resolution HSI from time $T1$ and high-spatial-resolution MSI from $T2$. In DAFN, the HSI is first upsampled, and features are extracted. Simultaneously, Gaussian blurring is applied to the MSI, followed by feature extraction. To address radiometric differences between the images, we propose a radiometric domain alignment module (RDAM) using a style transfer network. RDAM treats the HSI features as the style image and the MSI features as the content image, progressively aligning the radiometric distribution of MSI features with that of HSI features through convolution operations. To mitigate spatial geometric discrepancies, we introduce a spatial domain alignment module (SDAM), which calculates an affine transformation matrix between the HSI features and radiometrically aligned MSI features via convolution operations. The spatial geometry of the MSI features is then adjusted using grid generation and sampler interpolation. Next, we construct a change reconstruction module (CRM) to extract change information between the aligned features using convolutional operations and predict the low-spatial-resolution HSI at $T2$ by injecting change information into the upsampled $T1$ HSI. To further enhance spatial resolution, the feature extraction parameters, as well as the radiometric and spatial alignment parameters of the Gaussian-blurred MSI, are shared with the original MSI. A spatial reconstruction module (SRM) is then applied to fuse the predicted image with the aligned original MSI, ultimately producing the high-spatial-resolution HSI at $T2$. Notably, DAFN incorporates a supervised loss function between the predicted image and the ground truth, along with separate loss functions for radiometric and spatial alignment to continuously optimize the network. The main contributions of this article are as follows.

1) Considering the disparate imaging mechanisms and platforms between hyperspectral and multispectral sensors, which induce significant radiometric and spatial geometric discrepancies across multisource imagers, thus severely degrading temporal–spatial–spectral fusion quality, and the difficulty in obtaining prior image pairs at the observation time limits the practicality of the methods, we propose for the first time a simplified temporal–spatial–spectral fusion observation model for unpaired HSI-MSI. Building upon this framework, we design a Dual-DAFN for unpaired HSI-MSI fusion. This novel architecture simultaneously addresses cross-modal radiometric calibration and spatial geometric harmonization while mitigating data constraints, ultimately ensuring high-fidelity temporal–spatial–spectral fusion of HSI and MSI.
2) The DAFN framework integrates multiple dedicated network modules to collectively enhance fusion performance. First, we innovatively combine ResNet and Mamba architectures for joint local–global feature modeling, ensuring accurate extraction of multiscale spatial–spectral characteristics. Second, inspired by style transfer paradigms, we propose a novel RDAM to address cross-sensor radiometric inconsistencies and an SDAM to minimize geometric misalignment artifacts. Concurrently, the CRM and SRM are developed to facilitate hierarchical feature interaction and optimize spectral–spatial information fusion. Finally, a composite loss function, comprising radiometric domain alignment loss, spatial deformation regularization, and MSE constraints, is formulated to holistically guide network optimization.
3) DAFN preserves information in unchanged regions while accurately reconstructing information in changed regions. Experimental results on both simulated and real datasets demonstrate that DAFN achieves competitive fusion performance, outperforming six state-of-the-art methods in terms of spatial and spectral fidelity.

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The remainder of this article is organized as follows. Section II describes the relevant work. Section III provides a detailed description of the proposed DAFN. Section IV presents and analyzes the experimental results on simulated and real datasets. Finally, Section V concludes this article.

II. Related Work

A. Image Fusion Methods

Image fusion methods can be categorized into spatial–spectral fusion, temporal–spatial fusion, and temporal–spatial–spectral fusion, based on the specific resolution improvement target [16], [17]. Spatial–spectral fusion seeks to improve the spatial resolution of HSI by combining the spectral resolution of HSI with the spatial resolution of MSI, typically acquired from the same area at the same time [18]. Methods within this category include component substitution, multiresolution analysis, matrix decomposition, and deep learning-based approaches [19]. Component substitution methods extract the intensity component from the HSI and replace it with MSI to achieve spatial resolution enhancement [20], [21]. While these methods significantly enhance spatial resolution, they often introduce severe spectral distortion [8]. Multiresolution analysis techniques decompose images into multiple scales, injecting MSI details at different scales into HSI to improve spatial resolution [22], [23], [24]. These methods preserve spectral fidelity but tend to suffer from spatial distortion. Matrix decomposition methods [25], [26] leverage theories like mixed pixel decomposition, sparse representation, and tensor decomposition to separate and recombine spectral and spatial information from HSI and MSI, resulting in high-quality fusion. However, they are hindered by large parameter volumes, low computational efficiency, and limited robustness. Deep learning methods extract deep feature representations from both HSI and MSI, achieving fusion through nonlinear information recombination [27], [28]. These methods have demonstrated significant advantages in spatial–spectral fusion [29].

Temporal–spatial fusion aims to predict high-resolution (HR) images at a target time by utilizing pairs of HR and low-resolution (LR) images from different observation times, along with LR images at the target time for the same region [16]. Temporal–spatial fusion methods can be categorized into weight function methods, matrix decomposition methods, Bayesian methods, and deep learning methods [30], [31]. Weight function approaches predict missing pixel values by weighting auxiliary information [32]. Matrix decomposition methods decompose prior images into complex components and recombine them to predict HR images [33]. Bayesian methods transform temporal–spatial fusion into a maximum posterior probability estimation problem to achieve fusion [27], [34]. Deep learning methods use auxiliary information as input and predict missing pixel values through nonlinear operations, continuously optimizing network parameters by minimizing prediction errors [35], [36], [37], all of which have achieved notable success in temporal–spatial fusion.

Temporal–spatial–spectral fusion aims to predict HR HSI at missing time points by utilizing HSI and MSI from different times [1]. These methods are typically divided into iterative optimization methods and deep learning methods [38]. Iterative optimization approaches generally establish temporal–spatial–spectral transformation relationships between images using variational frameworks, obtaining fusion images through regularization constraints [39]. Notable examples include the integrated temporal–spatial–spectral fusion framework by Shen et al. [40]; a fusion framework based on semi-coupled sparse tensor decomposition by Peng et al. [41]; and a hybrid image fusion model proposed by Zhao and Huang [42]. However, these models often require the simultaneous use of EO-1 Hyperion, MODIS, and Landsat images, which can be limited by data availability. To address this, Huang et al. [43] proposed a Bayesian temporal–spatial–spectral fusion framework for the fusion of MODIS and Landsat images, while Sun et al. [38] and Zhou et al. [39] introduced mixed pixel decomposition-based temporal–spatial–spectral fusion methods for HSI and MSI in 2022 and 2023, respectively. Iterative optimization methods are often computationally expensive and suffer from low robustness [16]. Deep learning-based fusion methods, on the other hand, establish nonlinear mappings between auxiliary information and predicted values, continuously optimizing to refine fusion parameters [28], [44]. Examples of such methods include the multiscale Siamese convolutional neural network (CNN) for temporal–spatial–spectral fusion of HSI and MSI developed by Chen et al. [45]; the progressive spatial–temporal–spectral fusion network proposed by Chen et al. [46]; and the unsupervised 3-D tensor subspace decomposition network for HSI and MSI fusion introduced by Sun et al. [1].

B. Spectral Super-Resolution Methods

Spectral super-resolution methods have been widely studied in recent years [47], [48]. Arad et al. [49] developed a linear decomposition model in which they decomposed HSI into spectral bases and coefficients. They employed the K-SVD algorithm to learn the spectral bases from training data and then solved a coding problem based on the imaging model to obtain the corresponding coefficients. Li et al. [50] used a constrained sparse coding algorithm to reconstruct HSI by extracting a set of prototypes from the training data. Aeschbacher et al. [51] improved both performance and efficiency by employing an adjusted anchored neighborhood regression approach. Fu et al. [52] proposed a nonnegative sparse coding method to address the spectral super-resolution problem. Akhtar and Mian [53] applied Gaussian processes to model the self-similarity and spectral smoothness within HSI. Jia et al. [54] suggested that spectral signals lie within a low-dimensional subspace and introduced a principal manifold learning approach for spectral super-resolution. In recent years, deep learning has achieved significant advancements in spectral super-resolution. Nguyen et al. [55] introduced a radial basis function network for spectral super-resolution. Xiong et al. [56] used an upsampled red–green–blue (RGB) image as input to a CNN to predict HSI. Shi et al. [57]

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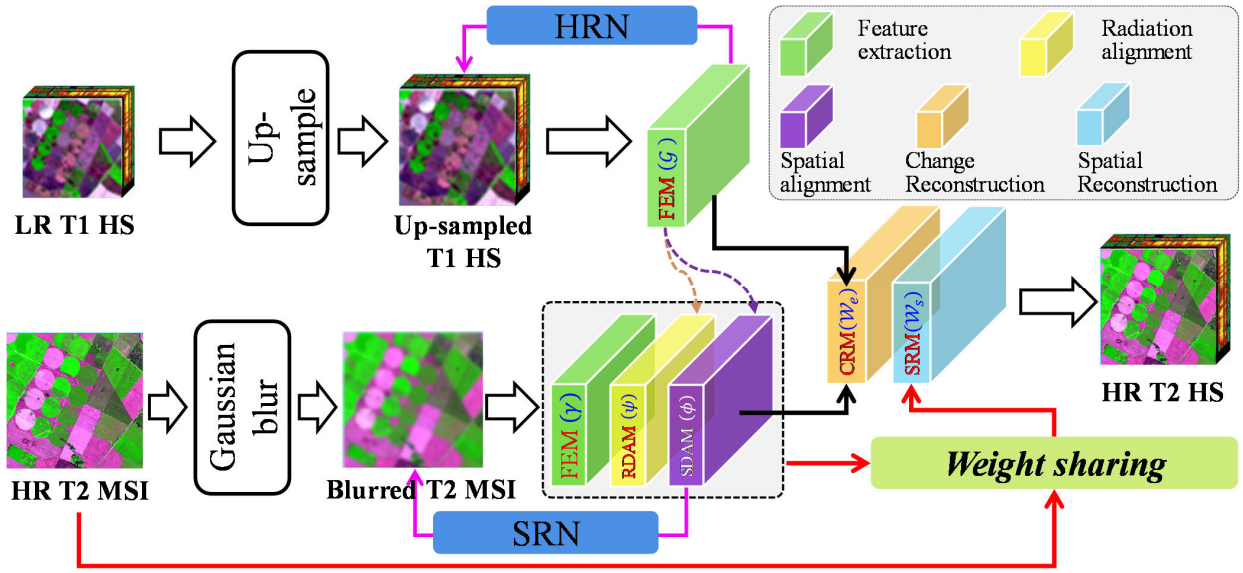


Fig. 3. DAFN.

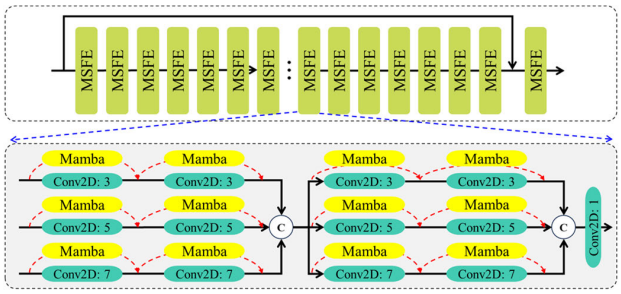


Fig. 4. FEM.

improved upon this approach by replacing manual upsampling with convolutional layers and further enhanced performance using advanced residual blocks and dense blocks. Additionally, Shi et al. [58] integrated a channel attention (CA) mechanism into the CNN to further boost learning performance. Alvarez-Gila et al. [59] proposed a conditional generative adversarial network (cGAN) for spectral super-resolution. To better leverage spatial–spectral contextual information, Zhu et al. [60] introduced a hybrid 2D-3D deep residual attention network combined with tensor constraints, designed for high-quality recovery of HSI from multitemporal RGB images, and further proposed a multitemporal spectral reconstruction network composed of several single-temporal spectral recovery networks [61]. Nie et al. [62] highlighted the importance of the spectral response function (SRF) of RGB images for spectral super-resolution and simultaneously learned both the SRF and the mapping function within the network. Fu et al. [63] introduced an SRF selection module to identify the optimal SRF and proposed a well-designed CNN to capture the spatial and spectral characteristics of HSI. Li et al. [60] designed a robust network with CA and incorporated the SRF in the loss function.

III. Methodology

A. Observation Model

In this article, we propose a dual-domain aligned temporal–spatial–spectral fusion observation model for HSI and MSI. This observation model effectively describes the critical relationship among LR HSI at $T1$, HR MSI at $T2$, and the target HR HSI at $T2$. The temporal–spatial–spectral fusion process aims to fuse an LR $\mathbf{X}\epsilon R^{w\times h\times L}$, where w and h represent the height and width of the LR $T1$ HSI, respectively, and L is the number of spectral bands, with an HR $T2$ MSI $\mathbf{Y}\epsilon R^{W\times H\times l}$ $(w < W, h < H, l < L)$ to obtain a target HR $T2$ HSI $\mathbf{Z}\epsilon R^{W\times H\times L}$. During the modeling process, $\mathbf{X}$ can be obtained by spatially degrading $\mathbf{Z}$ and subtracting the change information, which is expressed as follows:

$$\tilde{\mathbf{X}} = ((\mathbf{Z} * \mathcal{D}) - \mathbf{E}) - \mathbf{N}_x \tag{1}$$

where $\mathbf{E}$ represents the change information, $\mathcal{D}$ denotes the blurring operation, $\mathbf{N}_x\epsilon R^{W\times H\times L}$ is the noise term, and $\tilde{\mathbf{X}}$ indicates the spatially upsampled version of $\mathbf{X}$. By analyzing (1), it can be observed that $\tilde{\mathbf{X}}$ lacks both the change information $\mathbf{E}$ and the spatially blurred detail information $\mathbf{S}$ compared to $\mathbf{Z}$. Consequently, (1) can be reformulated as follows:

$$\tilde{\mathbf{X}} = \mathbf{Z} - \mathbf{E} - \mathbf{S} - \mathbf{N}_x. \tag{2}$$

Based on the temporal–spatial–spectral degradation model, a generalized supervised model for temporal–spatial–spectral fusion can be formulated as follows:

$$\arg\min_{\mathbf{E},\mathbf{S}} \left\|\mathbf{Z} - \left(\tilde{\mathbf{X}} + \mathbf{E} + \mathbf{S}\right)\right\|_F^2. \tag{3}$$

Therefore, accurately obtaining the change information $\mathbf{E}$ and the spatially blurred detail information $\mathbf{S}$ is crucial. In recent years, leveraging MSI as auxiliary information to predict the target HSI has proven to be an effective solution. However, due to the differing sensor design principles of MSI and HSI, significant radiometric discrepancies exist between the two datasets. Moreover, the varying imaging angles of satellites across different platforms result in substantial spatial misalignment of the same ground objects within the images. To address these issues, it is essential to perform radiometric domain alignment and spatial domain alignment on MSI. Therefore, $\mathbf{E}$ can be obtained as follows:

$$\mathbf{E} = \mathcal{W}_e\left(\phi\psi\left(\gamma\left(\tilde{\mathbf{Y}}\right)\right) - \mathcal{G}\left(\tilde{\mathbf{X}}\right)\right) \tag{4}$$

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where $\bar{\mathbf{Y}} \epsilon R^{W \times H \times l}$ represents the spatially blurred $\mathbf{Y}$; γ is the channel expansion matrix aimed at unifying the channel counts of $\mathbf{X}$ and $\mathbf{Y}$; and ψ denotes the radiometric domain alignment matrix, which will standardize the grayscale distributions of $\mathbf{X}$ and $\mathbf{Y}$. ϕ is the spatial domain alignment matrix, which will correct the spatial geometric information of $\mathbf{Y}$ within the model. $\mathcal{G}$ is the feature extraction matrix used to extract the features of $\tilde{\mathbf{X}}$, and $\mathcal{W}_e$ is the change gain matrix used to adjust the change features. Thus, the spatially blurred $\tilde{\mathbf{Z}}$ can be obtained as follows: $\tilde{\mathbf{Z}} = \tilde{\mathbf{X}} + \mathbf{E}$. To retrieve the detail information $\mathbf{S}$ and enhance the spatial resolution of $\tilde{\mathbf{Z}}$, $\mathbf{S}$ can be obtained as follows:

$$\mathbf{S} = \mathcal{W}_s\left(\phi\psi\left(\gamma(\mathbf{Y})\right) - \phi\psi\left(\gamma\left(\bar{\mathbf{Y}}\right)\right)\right) \tag{5}$$

where $\mathcal{W}_s$ is the spatial gain matrix used to adjust the spatial features. Finally, by combining (3)–(5), a generalized objective function for temporal–spatial–spectral fusion of HSI and MSI can be formulated as follows:

$$\arg\min_{\phi,\psi,\gamma} \Big\| \mathbf{Z} - \left(\tilde{\mathbf{X}} + \mathcal{W}_e\left(\phi\psi\left(\gamma\left(\bar{\mathbf{Y}}\right)\right) - \mathcal{G}\left(\tilde{\mathbf{X}}\right)\right) + \mathcal{W}_s\left(\phi\psi\left(\gamma(\mathbf{Y})\right) - \phi\psi\left(\gamma\left(\bar{\mathbf{Y}}\right)\right)\right)\right) \Big\|_F^2. \tag{6}$$

B. DAFN Overview

According to (6), $\bar{\mathbf{Y}}$ can be obtained by applying a Gaussian blur to $\mathbf{Y}$, which has been widely used in various studies. a technique that has been widely adopted in various studies. However, due to inconsistencies in the number of spectral channels between MSI and HSI, as well as significant radiometric and spatial differences caused by varying imaging mechanisms, the parameters $\mathcal{W}_e$, $\mathcal{W}_s$, γ, ψ, and ϕ must be determined in a nonlinear manner. Furthermore, cumulative errors make it challenging to accurately extract the change information $\mathbf{E}$ and detail information $\mathbf{S}$ through simple subtraction, necessitating the use of nonlinear operations for adaptive solutions.

In recent years, deep learning algorithms have demonstrated remarkable success in image reconstruction, owing to their powerful ability to solve complex nonlinear relationships. Therefore, based on deep learning principles, we propose DAFN. As illustrated in Fig. 3, $\mathbf{X}$ is first upsampled to obtain $\tilde{\mathbf{X}}$, and the MSI $\mathbf{Y}$ is processed with a Gaussian filter to produce $\bar{\mathbf{Y}}$. To address the channel inconsistency between $\tilde{\mathbf{X}}$ and $\bar{\mathbf{Y}}$, we construct a feature extraction module (FEM) to extract multiscale deep features from both $\tilde{\mathbf{X}}$ and $\bar{\mathbf{Y}}$, ensuring that the final number of feature channels is aligned (the final FEM parameters for $\bar{\mathbf{Y}}$ are denoted as γ). Next, we develop the RDAM to perform radiometric correction on the features of $\bar{\mathbf{Y}}$, using the features of $\tilde{\mathbf{X}}$ as guidance (where the RDAM parameters are denoted as ψ). To address spatial misalignment between the features of the two images, we construct the SDAM. SDAM aligns the features of $\bar{\mathbf{Y}}$ to those of $\tilde{\mathbf{X}}$, treating $\tilde{\mathbf{X}}$ features as the reference (the SDAM parameters are denoted as ϕ). Following alignment, we design the CRM, which takes the features of both $\tilde{\mathbf{X}}$ and the spatially and radiometrically aligned $\bar{\mathbf{Y}}$ as input to extract the change information (the CRM parameters are denoted as $\mathcal{W}_e$). The extracted change information is then injected into the features of $\tilde{\mathbf{X}}$, resulting in intermediate features $\tilde{\mathbf{Z}}$. To further enhance the spatial resolution of $\tilde{\mathbf{Z}}$, we share the FEM, RDAM, and SDAM parameters used for $\bar{\mathbf{Y}}$ with the original MSI $\mathbf{Y}$, obtaining its corrected features (due to $\mathbf{Y}$ not aligning its radiation and spatial domains with $\tilde{\mathbf{X}}$, weight sharing is necessary to weaken the radiometric and spatial differences between $\mathbf{Y}$ and $\tilde{\mathbf{X}}$ and achieve more accurate fusion results). Finally, we design an SRM to fuse the features of $\tilde{\mathbf{Z}}$ and the corrected $\mathbf{Y}$, producing the final fused $\hat{\mathbf{Z}}$ at the target time (the SRM parameters are denoted as $\mathcal{W}_s$).

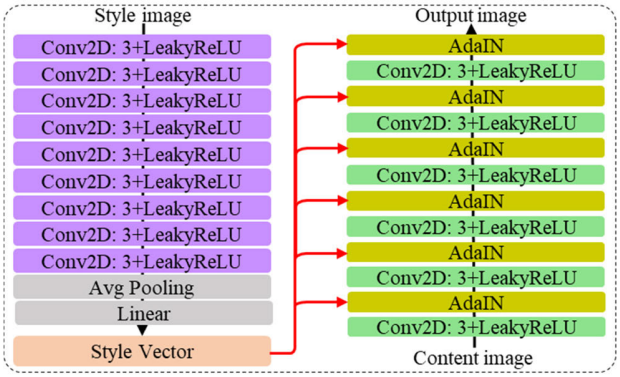


Fig. 5. RDAM.

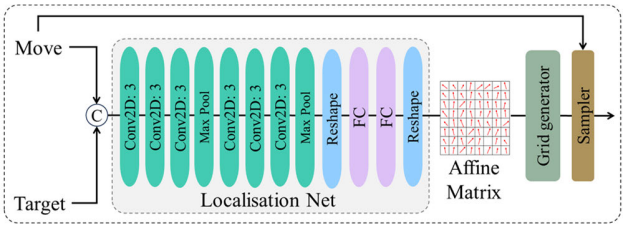


Fig. 6. SDAM.

It is worth noting that during our experiments, we constructed radiometric alignment loss functions, spatial alignment loss functions, and pixel-level loss functions to optimize the network. To ensure that the features extracted and aligned by the modules contain effective information, we further designed a hyperspectral reconstruction network (HRN) and a multispectral reconstruction network (MRN) to reconstruct the original HSI and MSI from their respective extracted features. Reconstruction loss functions were incorporated to jointly train DAFN, ultimately determining the optimal network parameters.

C. Network Branch Details

1) Feature Extraction Module: On the one hand, considering the inconsistency in the number of channels between HSI and MSI, and on the other hand, acknowledging that richer features contribute to more accurate feature reconstruction, we designed the FEM to extract features from both HSI and MSI. As shown in Fig. 4, FEM is a residual network primarily composed of multiscale feature extraction (MSFE) modules. Specifically, each MSFE consists of three branches, each containing convolution operations with kernels of varying sizes and Mamba blocks. This design is motivated by the fact that convolution kernels of different sizes capture different

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levels of image detail. Moreover, conventional convolution operations have a limited receptive field, capturing only local image features, which restricts the ability to extract global information. Given the recent success of the Mamba network in image processing and its superior computational efficiency over transformers for extracting global features, we integrate Mamba with convolution operations to simultaneously extract both multiscale global and local features from the images. For each MSFE, the multiscale global and local features of the input data are first extracted as follows:

$$\begin{cases} \mathcal{F}_3 = C_3(C_3(\mathbf{I}) + M(\mathbf{I})) + M(C_3(\mathbf{I}) + M(\mathbf{I})) \\ \mathcal{F}_5 = C_5(C_5(\mathbf{I}) + M(\mathbf{I})) + M(C_5(\mathbf{I}) + M(\mathbf{I})) \\ \mathcal{F}_7 = C_7(C_7(\mathbf{I}) + M(\mathbf{I})) + M(C_7(\mathbf{I}) + M(\mathbf{I})) \\ \mathcal{F}_I = \text{cat}(\mathcal{F}_3, \mathcal{F}_5, \mathcal{F}_7) \end{cases} \tag{7}$$

where $\mathbf{I} = [\tilde{\mathbf{X}}, \bar{\mathbf{Y}}]$ represents the input data; C_3, C_5, and C_7 denote convolution operations with 3×3, 5×5, and 7×7 kernels, respectively; M represents the Mamba operation; and F refers to the output features. Then, these features undergo multiscale fusion, and after feature stacking, the final image features are produced via a 1×1 convolution operation

$$\begin{cases} \mathcal{F}_{\mathcal{F}\mathbf{3}} = C_3(C_3(\mathcal{F}_\mathbf{I}) + M(\mathcal{F}_\mathbf{I})) + M(C_3(\mathcal{F}_\mathbf{I}) + M(\mathcal{F}_\mathbf{I})) \\ \mathcal{F}_{\mathcal{F}\mathbf{5}} = C_5(C_5(\mathcal{F}_\mathbf{I}) + M(\mathcal{F}_\mathbf{I})) + M(C_5(\mathcal{F}_\mathbf{I}) + M(\mathcal{F}_\mathbf{I})) \\ \mathcal{F}_{\mathcal{F}\mathbf{7}} = C_7(C_7(\mathcal{F}_\mathbf{I}) + M(\mathcal{F}_\mathbf{I})) + M(C_7(\mathcal{F}_\mathbf{I}) + M(\mathcal{F}_\mathbf{I})) \\ \mathcal{F}_{\mathcal{F}\mathbf{I}} = C_1(\text{cat}(\mathcal{F}_{\mathcal{F}\mathbf{3}}, \mathcal{F}_{\mathcal{F}\mathbf{5}}, \mathcal{F}_{\mathcal{F}\mathbf{7}})) \end{cases} \tag{8}$$

where C_1 denotes the convolution operation with a 1×1 kernel and $\mathcal{F}_{\mathcal{F}\mathbf{I}}$ represents the final image features. It is worth noting that in the experiments, FEM consists of five MSFE units. The corresponding calculation formula is expressed as follows:

$$\mathcal{F} = \mathcal{R}(\mathbf{I} + \mathcal{R}(\mathcal{R}(\mathcal{R}(\mathbf{I})))) \tag{9}$$

where $\mathcal{R}$ represents an MSFE operation.

2) Radiometric Domain Alignment Module: Due to differences in imaging mechanisms, HSI and MSI exhibit significant radiometric discrepancies, which substantially hinder the extraction of difference information between the images. Given that style transfer algorithms can approximate the grayscale distribution of a content image to that of a style image while preserving the detailed information of content image, we introduce the concept of style transfer and design the RDAM to mitigate the radiometric differences between HSI and MSI, as shown in Fig. 5. Specifically, the features of HSI are treated as the style image. After performing convolution operations, average pooling, and linear transformations, the style vector is obtained, which can be expressed as

$$w = l\left(\text{ap}\left(C_3^i\left(C_3^{i-1} \cdots \left(C_3^1(\mathcal{F}_\mathbf{X})\right)\right)\right)\right) \tag{10}$$

where i denotes the ith layer of convolution operations, $\mathcal{F}_\mathbf{X}$ represents the features of the style image, ap refers to the average pooling operation, l indicates the linear operation, and w signifies the style vector. Then, the SAR image is utilized as the content image, and style control is achieved through the AdaIN algorithm following the convolution operations:

$$\text{adaIN}(\mathcal{F}_\mathbf{Y}, w) = \sigma(w)\frac{\mathcal{F}_\mathbf{Y} - \mu(\mathcal{F}_\mathbf{Y})}{\sigma(\mathcal{F}_\mathbf{Y})} + \mu(w) \tag{11}$$

where $\mathcal{F}_\mathbf{Y}$ represents the features of MSI and μ and σ denote the mean and standard deviation, respectively. It is noteworthy that the RDAM employs nine layers of convolution operations when computing w, and a total of six convolution operations are set during the style transfer of $\mathcal{F}_\mathbf{Y}$. Additionally, the AdaIN algorithm is utilized for style control after convolution operations.

3) Spatial Domain Alignment Module: Multispectral and hyperspectral sensors operate at different imaging angles, resulting in significant spatial misalignment between the images. Direct extraction of the change information $\mathbf{E}$ and detail information $\mathbf{S}$ leads to substantial reconstruction errors. To address the spatial difference between the images, we developed the SDAM. As illustrated in Fig. 6, the SDAM consists of three submodules: the localization network, the grid generator, and the sampler. The primary objective of the localization network is to predict the parameters for the spatial transformation. The localization network outputs a set of parameters that define a spatial transformation, including translation, rotation, scaling, and other affine transformation parameters. In our experiments, the SDAM first stacks $\tilde{\mathbf{X}}$ and $\bar{\mathbf{Y}}$ along the channel dimension and then applies six layers of convolution operations to the stacked features. A max pooling module is inserted after every three convolution layers. After passing through a fully connected (FC) layer, a reshaping

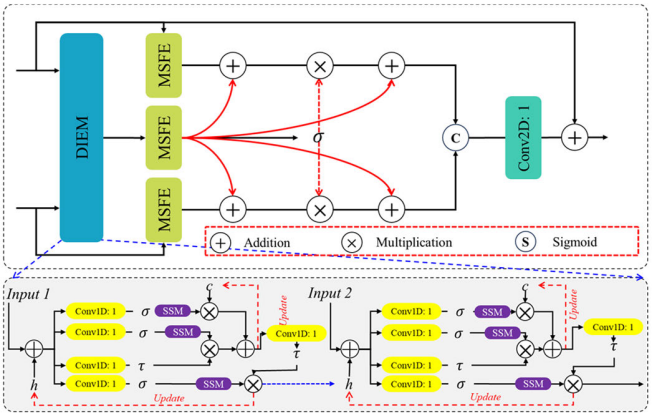


Fig. 7. CRM.

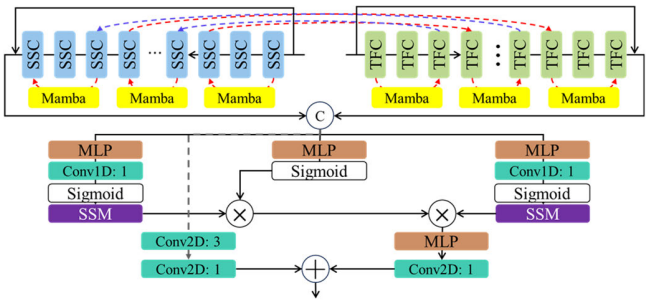


Fig. 8. SRM.

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operation is performed to obtain the final affine transformation matrix, which can be expressed as follows:

$$\begin{aligned}\mathcal{A}_\theta = \mathcal{U}(\mathcal{V}(\mathcal{V}(\mathcal{U}((\Omega(\Omega(\text{cat}(\mathcal{F}_{\mathbf{X}}, \mathcal{F}_{\mathbf{RY}})))))))) \\ \text{s.t. } \Omega \to \text{mp}\big(C_3^i\big(\cdots C_3^1(\text{cat}(\mathcal{F}_{\mathbf{X}}, \mathcal{F}_{\mathbf{RY}}))\big)\big)\end{aligned} \tag{12}$$

where $\mathcal{A}_\theta$ represents the affine transformation matrix, $\mathcal{F}_{\mathbf{RY}}$ denotes the features of $\bar{\mathbf{Y}}$ after radiometric domain alignment, $\mathcal{V}$ refers to the FC operation, $\mathcal{U}$ denotes the reshape operation, and mp represents the Max Pool operation. Following this, we define a grid generator, which is primarily used to receive the transformation parameters predicted by the localization network and generate a coordinate grid. This grid represents the new positions to which each pixel of the input image is mapped in the output image. The equation can be defined as

$$\begin{pmatrix} n_i^b \\ m_i^b \end{pmatrix} = \mathcal{A}_\theta(G_i) = \mathcal{A}_\theta \begin{pmatrix} n_i^f \\ m_i^f \\ 1 \end{pmatrix} = \begin{bmatrix} \theta_{11} & \theta_{12} & \theta_{13} \\ \theta_{21} & \theta_{22} & \theta_{23} \end{bmatrix} \begin{pmatrix} n_i^f \\ m_i^f \\ 1 \end{pmatrix} \tag{13}$$

where (n_i^b, m_i^b) represents the transformed pixel coordinates, (n_i^f, m_i^f) denotes the original pixel coordinates before transformation, G refers to the grid, and i indicates the ith pixel. Finally, we employed bilinear interpolation to construct the sampler, which samples pixel values from the input image based on the coordinate grid provided by the grid generator, thereby producing the transformed output image. Notably, in the experiments, SDAM treats $\mathcal{F}_{\mathbf{RY}}$ as the moving image and $\mathcal{F}_{\mathbf{X}}$ as the target image.

4) Change Reconstruction Module: After performing radiometric and spatial corrections on the MSI, it is essential to extract the change information and reconstruct $\tilde{\mathbf{Z}}$. In our experiments, we designed the CRM to accurately reconstruct $\tilde{\mathbf{Z}}$, as shown in Fig. 7. The CRM first introduces a difference information extraction module (DIEM), which takes $\mathcal{F}_{\mathbf{X}}$ and the spatially corrected $\mathcal{F}_{\mathbf{RY}}$ as inputs to obtain the initial change information. Subsequently, the MSFE module is employed to extract features from $\mathcal{F}_{\mathbf{X}}$, the spatially corrected $\mathcal{F}_{\mathbf{RY}}$, and the change information. A sigmoid activation function is then applied to analyze the distribution pattern of the change information. To refine the initial change information, we remove the change information from both $\mathcal{F}_{\mathbf{X}}$ and the spatially corrected $\mathcal{F}_{\mathbf{RY}}$ and extract the residual change information from the residual image based on the distribution pattern of the change information. This residual information is then added to the initial change information. Finally, the change information is stacked, and a 1×1 convolution operation is applied to obtain the final change information. This is then injected into $\mathcal{F}_{\mathbf{X}}$ to obtain the features of $\tilde{\mathbf{Z}}$. The specific equation can be expressed as

$$\begin{cases} \mathbf{E}_{\text{in}} = \mathcal{R}(\mathcal{G}(\mathcal{F}_{\mathbf{X}}, \mathcal{F}_{\mathbf{SRY}})) \\ \mathcal{F}_{\mathbf{Xe}} = (\mathcal{R}(\mathcal{F}_{\mathbf{X}}) + \mathbf{E}_{\text{in}}) \otimes \sigma(\mathbf{E}_{\text{in}}) + \mathbf{E}_{\text{in}} \\ \mathcal{F}_{\mathbf{SRYe}} = (\mathcal{R}(\mathcal{F}_{\mathbf{SRY}}) + \mathbf{E}_{\text{in}}) \otimes \sigma(\mathbf{E}_{\text{in}}) + \mathbf{E}_{\text{in}} \\ \tilde{\mathbf{Z}} = \mathcal{F}_{\mathbf{X}} + C_1(\text{cat}(\mathcal{F}_{\mathbf{Xe}}, \mathcal{F}_{\mathbf{SRYe}})) \end{cases} \tag{14}$$

where $\mathcal{G}$ represents the DIEM, $\mathcal{F}_{\mathbf{SRY}}$ refers to the spatially corrected $\mathcal{F}_{\mathbf{RY}}$, $\mathbf{E}_{\text{in}}$ denotes the initial change information, and

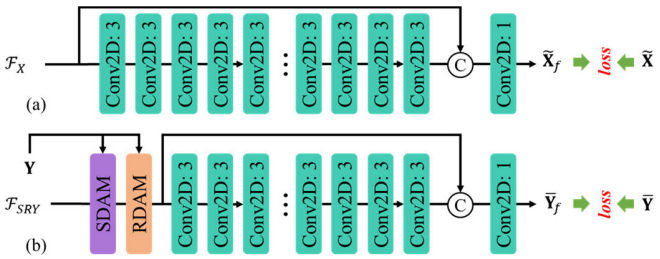


Fig. 9. (a) HRN and (b) MRN.

$\mathcal{F}_{\mathbf{Xe}}$ and $\mathcal{F}_{\mathbf{SRYe}}$ represent the optimized change information. It is important to note that $\mathcal{G}$ consists of a two-stage operation, primarily composed of two gated operations, and the gated operations constructed using the Mamba architecture. In the first stage, the features of the first image are input into the gated operation. After obtaining the hidden state and cell, the initial hidden state and cell are updated. The updated hidden state and cell are then used to replace the initial states for further network iteration and optimization. Upon completion of the iteration, the features of the second image, along with the hidden state and cell from the previous stage, are used as input for the second stage of iterative optimization. Through the gated operations, the DIEM continuously optimizes the input data and network parameters, facilitating an iterative process of error elimination. Each stage can be expressed as follows:

$$\begin{cases} f_t = \text{ssm}\big(\delta\big(wt_f \mathbf{F}_t + wt_{hf} \otimes H_{t-1} + wt_{cf} \odot C_{t-1} + b_f\big)\big) \\ i_t = \text{ssm}(\delta(wt_i \mathbf{F}_t + wt_{hi} \otimes H_{t-1} + wt_{ci} \odot C_{t-1} + b_i)) \\ C_{t-1} = f_t \odot C_{t-1} + i_t \odot \tanh \\ \qquad \otimes (wt_c \otimes \mathbf{F}_t + wt_{hc} \otimes H_{t-1} + b_c) \\ O_t = \text{ssm}(\delta(wt_o \otimes \mathbf{F}_t + wt_{ho} \otimes H_{t-1} + wt_{co} \odot C_{t-1} \\ \qquad + b_o)) \\ H_{t-1} = O_t \odot \tanh(C_{t-1}) \end{cases} \tag{15}$$

where wt represents the weight, b is the bias, $\mathbf{F}$ denotes the input features, H is the hidden output, C is the cell output, f refers to the forget gate, O is the output gate, δ represents the sigmoid function, tanh is the tanh function, t denotes the tth iteration, $\otimes$ represents the convolution operation, $\odot$ is the entry-wise product, and ssm stands for the state-space model.

5) Spatial Reconstruction Module: After obtaining $\tilde{\mathbf{Z}}$, enhancing its spatial resolution becomes crucial. Equation (6) utilizes the corrected $\mathbf{Y}$, with $\bar{\mathbf{Y}}$ subtracted, to extract spatial information, which is subsequently added to $\tilde{\mathbf{Z}}$ to yield the final $\mathbf{Z}$. However, this method may result in the loss of some spatial details, potentially leading to inaccurate image reconstruction. To ensure effective enhancement of spatial resolution, we first share the network parameters of FEM, RDAM, and SDAM with $\mathbf{Y}$ to derive the corrected $\mathbf{Y}$ ($\mathcal{F}_{\mathbf{SRHY}}$). Next, we construct the SRM [see Fig. 8], which directly takes $\mathcal{F}_{\mathbf{SRHY}}$ and $\tilde{\mathbf{Z}}$ as input to produce $\mathbf{Z}$. The SRM consists of two primary branches: one integrates a spatial–spectral convolutional block (SSC) with a Mamba block to extract features from $\tilde{\mathbf{Z}}$, while the other merges a texture feature convolutional (TFC) block with a Mamba block to extract features from

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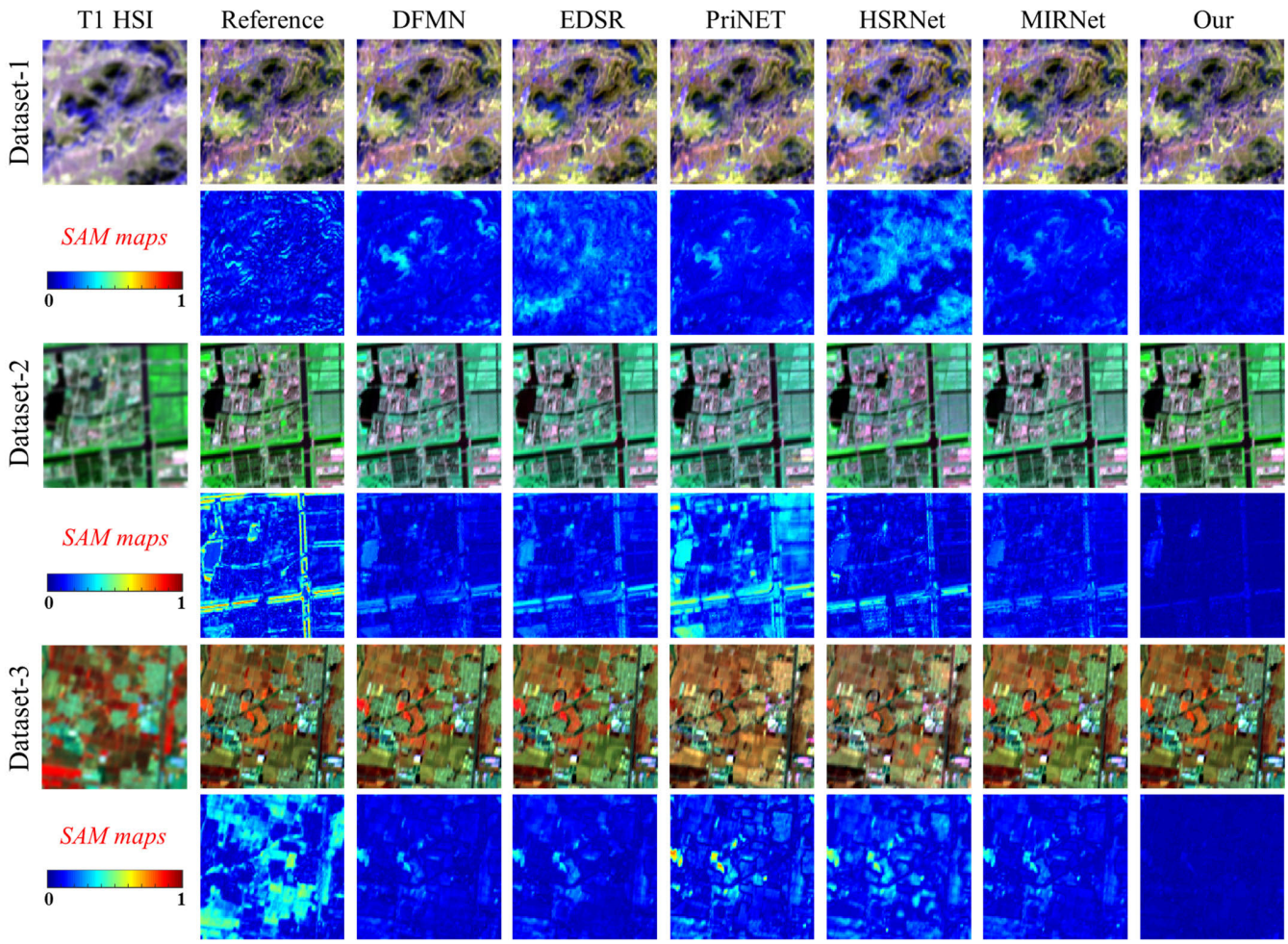


Fig. 10. Simulated dataset experimental results.

$\mathcal{F}_{\mathbf{SRHY}}$. Furthermore, a feature interaction branch is included between the SSC block and the TFC block to improve feature fusion. The specific equation can be expressed as follows:

$$\begin{cases} \mathcal{F}_{\text{ss}}^{1} = \mathcal{G}(\mathcal{F}_{\mathbf{SRHY}}) \\ \mathcal{F}_{\text{tf}}^{1} = \mathcal{H}(\tilde{\mathbf{Z}}) \\ \mathcal{F}_{\mathbf{ss}}^{\mathbf{2}} = \mathcal{G}(\mathcal{F}_{\text{ss}}^{1} + \mathcal{F}_{\text{tf}}^{1}) \\ \mathcal{F}_{\text{tf}}^{2} = \mathcal{H}(\mathcal{F}_{\text{ss}}^{1} + \mathcal{F}_{\text{tf}}^{1}) \\ \quad \cdots \\ \mathcal{F}_{\text{ss}}^{i} = \mathcal{G}(\mathcal{F}_{\text{ss}}^{i-1} + \mathcal{F}_{\text{tf}}^{i-1}) + \mathcal{F}_{\mathbf{SRHY}} \\ \mathcal{F}_{\text{tf}}^{i} = \mathcal{H}(\mathcal{F}_{\text{ss}}^{i-1} + \mathcal{F}_{\text{tf}}^{i-1}) + \tilde{\mathbf{Z}} \end{cases} \tag{16}$$

$$\begin{aligned} \mathcal{G} &\rightarrow C_3(C_3(C_3(\mathbf{IN})) + M(C_3(\mathbf{IN}))) \\ \mathcal{H} &\rightarrow \text{CD}_3(\text{CD}_3(\text{CD}_3(\mathbf{IN})) + M(\text{CD}_3(\mathbf{IN}))) \end{aligned} \tag{17}$$

where $\mathcal{G}$ represents the spatio-spectral convolutional block, which consists of three convolutional layers; $\mathcal{H}$ refers to the TFC block, composed of three central difference convolutional layers; $\mathcal{F}_{\text{ss}}$ denotes the spatio-spectral features; $\mathcal{F}_{\text{tf}}$ represents the texture features; and **IN** stands for the input data. After extracting features from $\mathcal{F}_{\mathbf{SRHY}}$ and $\tilde{\mathbf{Z}}$, these features are stacked to obtain $\mathcal{F}_{\text{st}}$. We then designed a three-branch network by combining convolution with Mamba to further extract features from $\mathcal{F}_{\text{st}}$

$$\begin{cases} \mathbf{Q} = \text{ssm}(\delta(C_1(\text{mlp}(\mathcal{F}_{\text{st}})))) \\ \mathbf{K} = \delta(\text{mlp}(\mathcal{F}_{\text{st}})) \\ \mathbf{V} = \text{ssm}(\delta(C_1(\text{mlp}(\mathcal{F}_{\text{st}})))) \end{cases} \tag{18}$$

where **Q**, **K**, and **V** represent the output features and mlp refers to the multilayer perceptron. Afterward, **Q** and **K** undergo a dot product, which is then multiplied by **V** to obtain the attention features

$$\mathbf{A} = \mathbf{V} \odot ((\mathbf{Q} \odot \mathbf{K})) \tag{19}$$

where **A** represents the attention features. Finally, the attention features are processed through the mlp and convolution operations and then added to the convolutional features of $\mathcal{F}_{\text{st}}$ to obtain the final $\hat{\mathbf{Z}}$

$$\hat{\mathbf{Z}} = C_1(C_3(\mathcal{F}_{\text{st}})) + C_1(\text{mlp}(\mathbf{A})). \tag{20}$$

6) HRN and MRN: Considering that effective $\mathcal{F}_{\mathbf{X}}$ and $\mathcal{F}_{\mathbf{SRY}}$ can reconstruct the original $\tilde{\mathbf{X}}$ and $\tilde{\mathbf{Y}}$, we constructed the HRN and SRN to ensure that $\mathcal{F}_{\mathbf{X}}$ and $\mathcal{F}_{\mathbf{SRY}}$ contain valid information for subsequent image reconstruction [see Fig. 9]. Specifically, we used $\mathcal{F}_{\mathbf{X}}$ and $\mathcal{F}_{\mathbf{SRY}}$ as input data to reconstruct $\tilde{\mathbf{X}}$ and $\tilde{\mathbf{Y}}$. The HRN is primarily composed of a ResNet architecture, utilizing a total of 16 ResNet modules to

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reconstruct $\tilde{\mathbf{X}}$. Similarly, the SRN also consists of 16 ResNet modules. Unlike the HRN, the SRN incorporates both SDAM and RDAM modules, utilizing $\mathbf{Y}$ as the style image and spatial reference image to calibrate $\mathcal{F}_{\mathbf{SRY}}$ for the reconstruction of $\tilde{\mathbf{Y}}$.

D. Joint Training

To effectively train DAFN and obtain high-quality fused images, we designed the following loss function for network parameter optimization:

$$\mathcal{L} = \mathcal{L}_{\text{da}} + \mathcal{L}_{\text{sa}} + \mathcal{L}_{\text{mse}} \tag{21}$$

where L represents the overall loss. $\mathcal{L}_{\text{da}}$ denotes the radiometric alignment loss function, $\mathcal{L}_{\text{sa}}$ is the spatial alignment loss function, and $\mathcal{L}_{\text{mse}}$ represents the mse loss between the fused result and the ground truth. The radiometric alignment loss $\mathcal{L}_{\text{da}}$ is defined as follows:

$$\mathcal{L}_{\text{da}} = \frac{1}{4K^2}\left\|\mathbf{C}_{\mathcal{F}_{\mathbf{X}}} - \mathbf{C}_{\mathcal{F}_{\mathbf{SY}}}\right\|_F^2 \tag{22}$$

where K represents the number of spectral bands, $\mathbf{C}_{\mathcal{F}_{\mathbf{X}}}$ denotes the covariance matrix of $\mathbf{C}_{\mathcal{F}_{\mathbf{SY}}}$ represents the radiometrically aligned $\tilde{\mathbf{Y}}$, and $\mathcal{F}_{\mathbf{SY}}$ denotes the covariance matrix of $\mathcal{F}_{\mathbf{SY}}$. Furthermore, $\mathbf{C}_{\mathcal{F}_{\mathbf{SY}}}$ can be expressed as follows:

$$\begin{cases} \mathbf{C}_{\mathcal{F}_{\mathbf{X}}} = \dfrac{1}{n-1}\left(\mathcal{F}_{\mathbf{X}}^T\mathcal{F}_{\mathbf{X}} - \dfrac{1}{n}\left(1^T\mathcal{F}_{\mathbf{X}}\right)^T\left(1^T\mathcal{F}_{\mathbf{X}}\right)\right) \\ \mathbf{C}_{\mathcal{F}_{\mathbf{SY}}} = \dfrac{1}{n-1}\left(\mathcal{F}_{\mathbf{SY}}^T\mathcal{F}_{\mathbf{SY}} - \dfrac{1}{n}\left(1^T\mathcal{F}_{\mathbf{SY}}\right)^T\left(1^T\mathcal{F}_{\mathbf{SY}}\right)\right). \end{cases} \tag{23}$$

To reduce the spatial geometric errors between images, we also designed the NCC loss function as $\mathcal{L}_{\text{sa}}$, which is specifically expressed as follows:

$$\mathcal{L}_{\text{sa}} = \frac{\sum\left(\mathcal{F}_{\mathbf{X}} - \mu_{\mathcal{F}_{\mathbf{X}}}\right)\left(\mathcal{F}_{\mathbf{SY}} - \mu_{\mathcal{F}_{\mathbf{SY}}}\right)}{\sqrt{\sum\left(\mathcal{F}_{\mathbf{X}} - \mu_{\mathcal{F}_{\mathbf{X}}}\right)^2\sum\left(\mathcal{F}_{\mathbf{SY}} - \mu_{\mathcal{F}_{\mathbf{SY}}}\right)^2}} \tag{24}$$

where $\mu_{\mathcal{F}_{\mathbf{X}}}$ and $\mu_{\mathcal{F}_{\mathbf{SY}}}$ represent the means of $\mathcal{F}_{\mathbf{X}}$ and $\mathcal{F}_{\mathbf{SY}}$, respectively. Ultimately, we designed the MSE loss function as $\mathcal{L}_{\text{mse}}$ to impose constraints on the fusion results

$$\mathcal{L}_{\text{mse}} = \frac{1}{n}\sum_{i=1}^{n}\left(\mathbf{Z}_i - \hat{\mathbf{Z}}_i\right). \tag{25}$$

IV. Experimental Results and Analysis

A. Experimental Dataset

In the experiments, all HSIs were acquired from the ZY-1 02-D satellite sensor. Each hyperspectral scene was captured across 166 spectral bands within the 400–2500-nm range, with a spatial resolution of 30 m. After removing bad bands, a total of 100 bands were used in the experiments. For the simulated data, HSIs were obtained from three regions, with each region containing two scenes captured at different times. The time intervals between these scenes varied as follows: Dataset 1 (June 5, 2021, and December 19, 2021), Dataset 2 (December 16, 2020, and December 19, 2021), and Dataset 3 (November 12, 2019, and February 6, 2021). The HSIs from the $T1$ period were processed with Gaussian blur and downsampled by a factor of 4 to obtain $\mathbf{X}$, while the HSIs from the $T2$ period were spectrally downsampled using the S2A SRF to obtain $\mathbf{Y}$. The original HSIs from $T2$ were used as the ground truth for network training. During data cropping, a spatial stride of 64 was applied, resulting in 6134 patches per region, each with a size of 128 × 128 pixels. In the experiments, datasets from the three regions were combined to train the DAFN, using 15 000 patches for training and 3402 patches for validation.

For the real dataset, MSIs were captured by the Sentinel-2 sensor with a spatial resolution of 10 m. In the experiments, the real data underwent a downscaling simulation, and the pretrained DAFN model was fine-tuned using the simulated data. After training, the network was applied to the real HSI from the $T1$ period and MSI from the $T2$ period to generate the fusion result for the real dataset. The real dataset included the following: Dataset 1 ($T1$ HSI-December 11, 2020, $T2$ MSI-October 15, 2021) and Dataset 2 ($T1$ HSI-February 8, 2021, $T2$ MSI-December 24, 2021).

B. Experimental Setting

Our DAFN is the first method to perform temporal, spatial, and spectral fusion of MSI and HSI without relying on paired images, and as such, no directly comparable algorithms exist. In our experiments, we selected five spectral super-resolution methods for comparative analysis: deep function mixture network (DFMN) [64], HSRNet [65], PriNET [66], EDSR [67], and MIRNet [68].

Additionally, five widely used metrics were employed to assess the fusion performance of DAFN on the simulated dataset: peak signal-to-noise ratio (PSNR), spectral angle mapper (SAM), Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS), structural similarity index measure (SSIM), and cross correlation (CC) [16], [69]. Higher PSNR values indicate better image quality, while ideal SAM and ERGAS values are 0, with larger values reflecting worse image quality. For SSIM and CC, the ideal value is 1, with smaller values indicating more severe spatial–spectral distortions.

To train DAFN, the Adam optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.99$ was used, with the learning rate set to $1e^{-4}$ and a batch size of 4. The training was conducted on an Ubuntu 22.04 computer equipped with 128 GB of CPU memory and a GeForce GTX 4090 D GPU.

C. Experimental Results

1) Simulated Dataset's Experimental Results: Fig. 10 shows the experimental results on the simulated datasets, along with the corresponding SAM maps. Simulated Dataset-1 primarily highlights the transition from bare soil to vegetation. Due to the preservation of spatial information, all spectral super-resolution methods exhibit clear spatial details. However, some spectral distortions persist in the fused results. For instance, DFMN shows significant spectral distortions in shadowed regions, EDSR exhibits spectral reconstruction errors in sloped areas, and HSRNet experiences global spectral distortion. In contrast, both PriNET and MIRNet demonstrate better spectral reconstruction accuracy, while our method closely aligns with the reference image, exhibiting minimal

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TABLE I
QUANTITATIVE EVALUATION OF EXPERIMENT RESULTS

Methods	Dataset-1				Dataset-2				Dataset-3			
	SAM	ERGAS	PSNR	SSIM	SAM	ERGAS	PSNR	SSIM	SAM	ERGAS	PSNR	SSIM
DFMN	5.77	13.61	28.49	0.931	5.01	22.16	22.03	0.856	6.95	16.53	25.31	0.843
EDSR	7.21	15.83	26.85	0.911	5.30	21.67	22.16	0.858	9.81	30.94	22.67	0.825
PriNET	12.27	39.74	14.33	0.742	7.55	51.05	16.58	0.701	6.97	14.76	26.09	0.807
HSRnet	6.50	43.56	19.62	0.81	8.12	19.34	22.14	0.713	9.42	19.88	21.51	0.758
MIRNet	6.32	15.45	26.63	0.917	6.06	34.28	19.14	0.817	6.93	14.62	26.26	0.841
DAFN	**2.02**	**4.36**	**39.01**	**0.989**	**2.15**	**4.05**	**38.28**	**0.971**	**5.27**	**10.95**	**29.11**	**0.920**

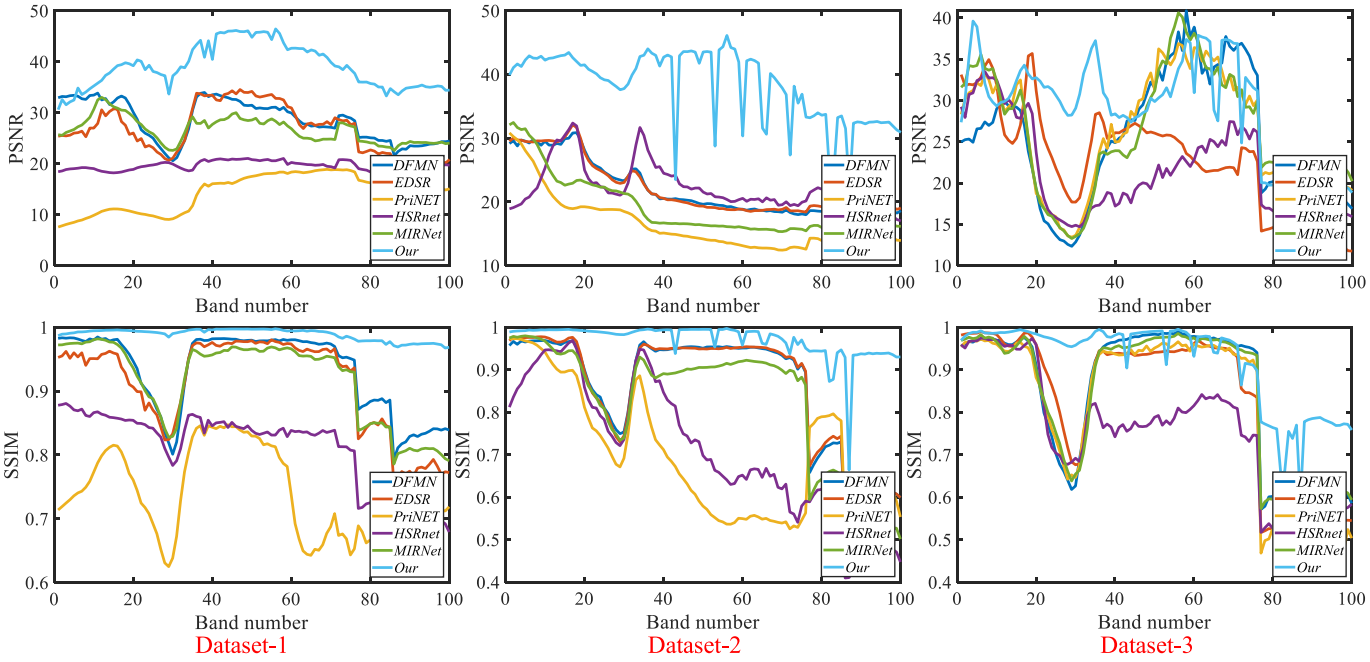


Fig. 11. Comparison of experimental results of PSNR and SSIM for each band.

spectral distortion and a noticeable enhancement in spatial resolution.

In Simulated Dataset-2, the spectral differences between images captured on different dates are more pronounced than in Dataset-1, reflecting changes in vegetation phenology and building distributions. All spectral super-resolution methods show notable radiometric discrepancies from the reference image in vegetation areas, particularly PriNET. Our method achieves the best visual results and the lowest spectral reconstruction error.

Simulated Dataset-3 exhibits the most severe land-cover changes, highlighting transformations in vegetation and building areas within farmland. While all benchmark methods preserve clear spatial details, they exhibit significant spectral reconstruction errors, particularly PriNET and HSRNet, especially in vegetation areas. Our method not only preserves clear spatial details but also significantly reduces spectral reconstruction errors compared to other methods, achieving the best performance.

To ensure a fair comparison among all methods, Table I shows the quantitative evaluation results of the fused results. In Simulated Dataset-1, PriNET and EDSR yield the poorest results across all metrics. DFMN, HSRNet, and MIRNet exhibit similar SAM, PSNR, and SSIM values; however, the ERGAS value for HSRNet is significantly larger than that of MIRNet, indicating poorer fusion performance. Our proposed DAFN achieves the best results across all metrics. In Simulated Dataset-2, all benchmark methods yield SAM $>$ 5, ERGAS $>$ 18, PSNR $<$ 23, and SSIM $<$ 0.86. HSRNet shows the worst performance, with the poorest SAM and ERGAS values, followed by PriNET. DFMN, EDSR, and MIRNet produce similar evaluation results. DAFN again demonstrates the best performance across all metrics in Simulated Dataset-2. In Simulated Dataset-3, EDSR exhibits the worst SAM and ERGAS values, while HSRNet shows the lowest PSNR and SSIM. DAFN consistently delivers the best results across all metrics.

To evaluate the reconstruction performance for each band, Fig. 11 shows the PSNR and SSIM values for the fused results across all bands. It is evident that DAFN outperforms the benchmark methods in nearly all bands.

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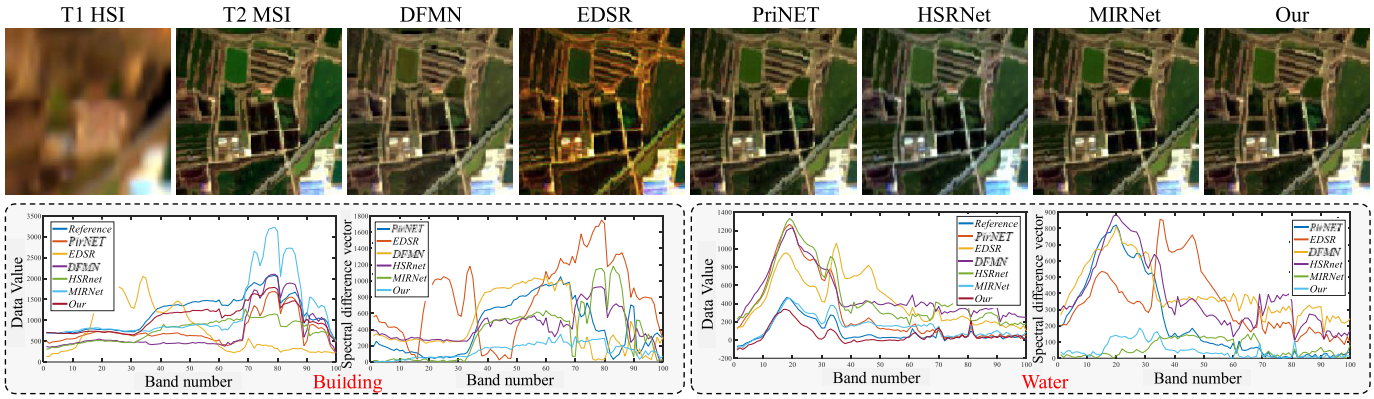


Fig. 12. Comparison of experiment results of real dataset-1.

TABLE II
QUANTITATIVE EVALUATION RESULTS OF ABLATION EXPERIMENT

HRN and SRN	RDAM	SDAM	Mamba	$\mathcal{L}_{da}$	$\mathcal{L}_{sa}$	$\mathcal{L}_{mse}$	SAM	PSNR
×	√	√	√	√	√	√	7.82	26.58
√	×	√	√	√	√	√	10.63	22.99
√	√	×	√	√	√	√	8.05	23.67
√	√	√	×	√	√	√	7.22	26.97
√	√	√	√	×	√	√	11.90	21.07
√	√	√	√	√	×	√	7.99	24.31
√	√	√	√	√	√	×	13.67	18.33
√	√	√	√	√	√	√	5.27	29.11

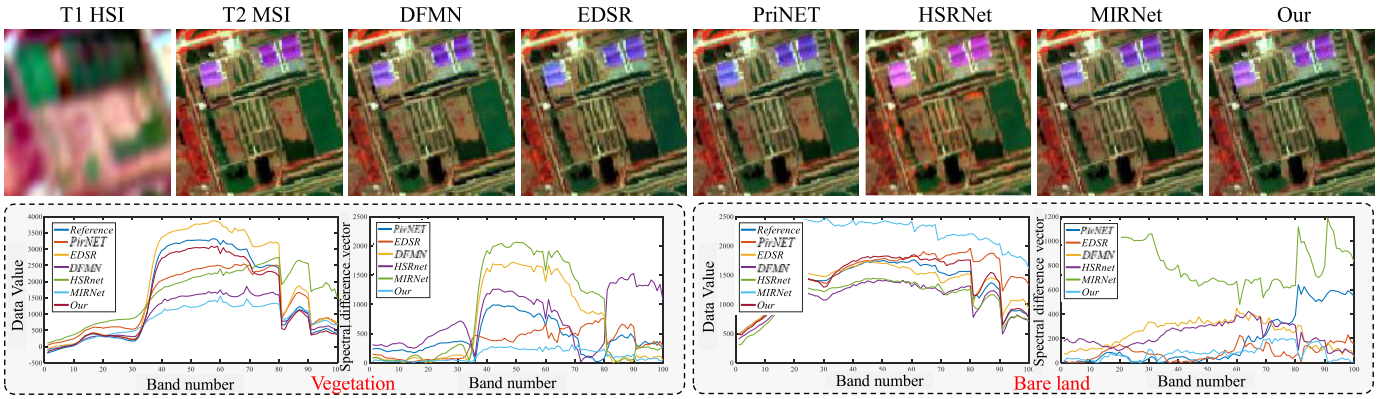


Fig. 13. Comparison of experiment results of real dataset-2.

TABLE III
COMPARATIVE ANALYSIS OF COMPUTATIONAL COMPLEXITY

Running information	DFMN	EDSR	PriNET	HSRnet	MIRNet	DAFN
Param (M)	4.23	2.31	5.10	2.95	3.01	2.87
FLOPS (G)	6.91	3.39	6.28	4.63	5.78	3.69
Time (s)	1.96	1.52	2.03	1.64	1.86	1.35
Software Platform	Python	Python	Python	Python	Python	Python

2) Real Dataset's Experimental Results: To evaluate the performance of DAFN on real datasets, we conduct comparative experiments on two real-world datasets. Real Dataset-1 captures changes in water bodies, and Fig. 12 presents the experimental results for various methods. All methods preserve clear spatial details. However, EDSR and HSRNet exhibit significant global spectral distortions, whereas other methods demonstrate more consistent radiometric information. The radiometric reconstruction results from the other methods are similar. To ensure a fair comparison of the fusion results, we selected spectral curves from invariant ground objects, such as vegetation and bare soil, as references and compared them

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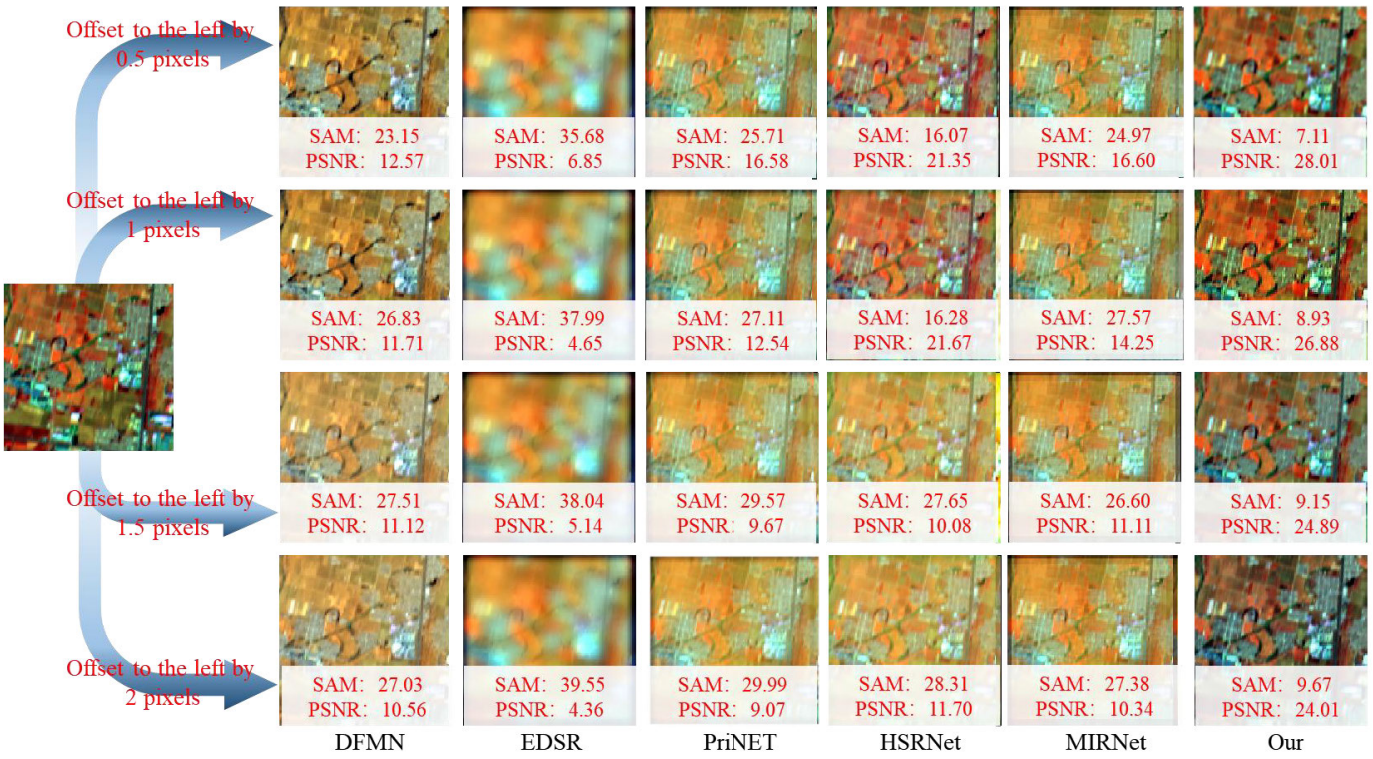


Fig. 14. Experimental results of different methods on different offset datasets.

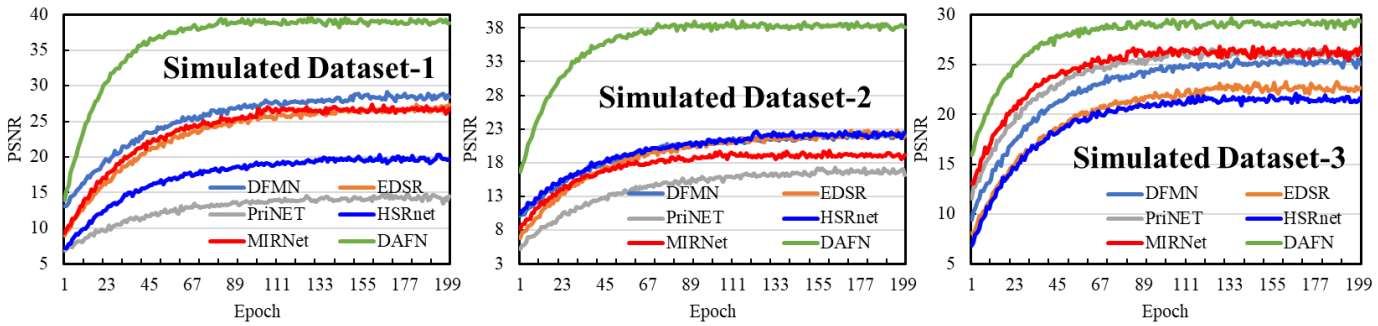


Fig. 15. Comparison results of convergence rates of different networks.

with the spectral curves obtained using different methods. The spectral curve comparison is shown at the bottom left of Fig. 12, while the curve difference vector plots are displayed on the right (where values closer to zero indicate better fusion results). In the building spectral curve comparison, EDSR shows pronounced errors in both peak and valley points, with its spectral curve being entirely inconsistent with the reference. The trends of the spectral curves from other methods are consistent with the reference, though notable differences in the amplitude of fluctuations are observed. For the water body curve, all methods exhibit trends consistent with the reference curve. However, EDSR shows erroneous peaks and valleys between bands 30 and 50. Our proposed DAFN method achieves results closest to the reference curves for both building and water body spectral curves.

Real Dataset-2 primarily reflects changes in building and vegetation areas, with the experimental results shown in Fig. 13. All methods maintain clear spatial details, but the color in the building areas from HSRNet noticeably differs from the other fusion results. The spectral curve comparison results and curve difference vector plots are shown in the bottom row of Fig. 13. In the vegetation spectral curve comparison, all methods follow the trend of reference curve, with PriNET and MIRNet exhibiting the most significant fluctuations. EDSR produces the best vegetation spectral curve results among the benchmark methods. For the bare soil spectral curve, the result of MIRNet is entirely inconsistent with the reference curve, while the other benchmark methods closely followed the reference trend and values. The spectral curves produced by DAFN for both vegetation and bare soil are the most consistent with the reference curves.

3) Ablation Experiment: To assess the contributions of different network modules for DAFN, we conduct a series of ablation experiments on Simulated Dataset-3. The ablation experiments include the removal of HRN and SRN, RDAM, SDAM, Mamba, and the loss functions. Notably, all

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ablation experiments are conducted with a consistent setting of 200 epochs to ensure a fair comparison. The quantitative results of these ablation experiments are presented in Table II. HRN and SRN are employed to constrain the extracted feature values, and their effectiveness is demonstrated in the experiments. When HRN and SRN are ablated, the SAM increased by 2.55 and the PSNR decreased by 2.53. RDAM primarily facilitates the radiometric alignment of the images. When RDAM is removed, SAM increased by 5.36 and PSNR decreased by 6.12, indicating that reducing radiometric differences significantly enhances fusion quality. SDAM, which is integrated into DAFN for spatial alignment, contributes to a reduction in SAM by 2.78 and an increase in PSNR by 5.44. To enable efficient long-range modeling, the Mamba network is incorporated into DAFN. We perform an ablation of Mamba to assess its importance. When Mamba is removed, SAM increased by 1.95 and PSNR decreased by 2.14. Furthermore, the design of the loss functions is critical to improving network performance. We conduct ablation experiments on the loss functions $\mathcal{L}_{\text{da}}$, $\mathcal{L}_{\text{sa}}$, and $\mathcal{L}_{\text{mse}}$ individually. The results show that the combination of $\mathcal{L}_{\text{da}}$, $\mathcal{L}_{\text{sa}}$, and $\mathcal{L}_{\text{mse}}$ significantly enhances the network performance, with $\mathcal{L}_{\text{mse}}$ being the most important, followed by $\mathcal{L}_{\text{da}}$. In summary, each module within DAFN contributes effectively to improving fusion performance.

Additionally, we compare the performance of various methods under varying degrees of spatial shifts between images. To ensure a fair comparison, pixel shifts are applied to all samples within the range of [0.5–2], and all networks are fine-tuned on the shifted datasets for 50 epochs based on their pretrained models. Fig. 14 presents the results for different methods when images are shifted by 0.5, 1, 1.5, and 2 pixels. It is evident that due to the lack of prior information, spectral super-resolution methods exhibit significant spatial and spectral distortions. EDSR produces the worst results, while HSRNet performs relatively well when the shift is within the range [0–1] but experiences significant reconstruction errors when the shift exceeds 1 pixel. Other methods struggle to effectively eliminate spatial misalignment, resulting in noticeable spectral distortions. In contrast, DAFN, with its ability to efficiently align images spatially, achieves the best results across datasets with varying degrees of spatial shifts, although fusion performance declines as the magnitude of the shift increased.

Fig. 15 illustrates the convergence rates of different methods. In Simulated Dataset-1, PriNET and HSRnet demonstrate rapid PSNR improvements during 1–110 epochs followed by stabilization, while EDSR and MIRNet exhibit initial PSNR values of approximately 8 with rapid enhancement in 1–90 epochs. DFMN and DAFN achieve the highest initial PSNR values, where DAFN converges around 70 epochs, whereas DFMN shows slower convergence at approximately 90 epochs. In Simulated Dataset-2, all baseline methods converge between 110 and 130 epochs. Notably, PriNET presents the lowest initial PSNR, EDSR and MIRNet share comparable initial values, and DFMN/HSRnet show initial values around 10. DAFN attains the fastest convergence rate, stabilizing after 90 epochs, with the highest initial PSNR. For Simulated Dataset-3, HSRnet and EDSR start at six PSNRs with steep increases until 120 epochs. DFMN, MIRNet, and PriNET converge around 100 epochs, while DAFN consistently maintains the fastest convergence rate and highest initial PSNR.

Table III presents a comparative analysis of computational complexity across six methods, with DAFN exhibiting competitive efficiency in parameter count (2.87 M), outperforming DFMN (4.23 M), PriNET (5.10 M), and MIRNet (3.01 M) while slightly exceeding EDSR (2.31 M) and HSRnet (2.95 M). In terms of computational load, DAFN achieves 3.69 G FLOPS, demonstrating superior efficiency compared to DFMN (6.91 G), PriNET (6.28 G), MIRNet (5.78 G), and HSRnet (4.63 G), though marginally higher than EDSR (3.39 G). Notably, DAFN attains the fastest inference time (1.35 s), surpassing all baselines, including DFMN (1.96 s), PriNET (2.03 s), MIRNet (1.86 s), HSRnet (1.64 s), and EDSR (1.52 s), thereby striking an optimal balance between model compactness, computational efficiency, and runtime performance under a unified Python framework.

V. Conclusion

This article proposes a Dual-DAFN for unpaired hyperspectral and MSI fusion. DAFN does not require paired HSI and MSI acquired on the same date as prior information. By integrating CNNs and Mamba, DAFN effectively models both long- and short-range dependencies in the images. Additionally, the design of RDAM and SDAM effectively mitigates radiometric and spatial discrepancies between sensors. A carefully designed loss function enables joint training of the network. Through extensive experimentation, we conclude that DAFN outperforms state-of-the-art spectral super-resolution methods and demonstrates superior robustness. Furthermore, the distinct modules within DAFN collectively contribute to enhanced fusion performance, validating the rationality of the network design. Additionally, DAFN exhibits strong adaptability to various spatial registration errors, effectively mitigating spatial misalignment between images. However, Param and FLOPS of DAFN are not yet optimized; future work will focus on refining the network architecture by integrating knowledge distillation to reduce computational complexity while maintaining fusion efficacy.

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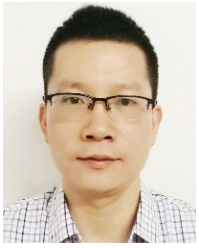
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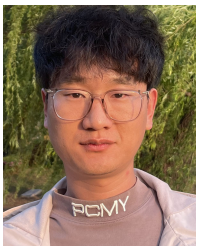
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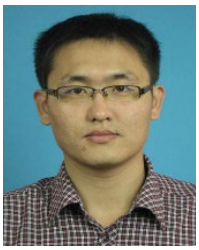
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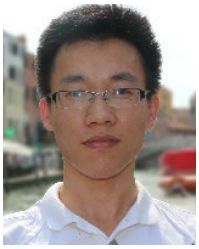
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